

Individual Assignment : Week 4

January 25, 2026

This assignment must be solved and submitted **individually** and you are not allowed to work in groups. This assignment count towards your final grade and have to be submitted in order to pass the course. You must follow the report guidelines found in `guidelines.pdf`. The assignment contains questions on the topics we have covered in the course up to now including this week. The page limit for this assignment is **10 pages** including everything, i.e. illustrations and code snippets.

1. Image filtering

- 1.1. (1 point) Consider the following finite difference approximations of the image derivatives given by:

$$\frac{\partial I}{\partial x}(x, y) \approx \frac{I(x+1, y) - I(x-1, y)}{2}$$
$$\frac{\partial I}{\partial y}(x, y) \approx \frac{I(x, y+1) - I(x, y-1)}{2}$$

Deliverables: Assuming that we filter an image using correlation, what are the filter kernels that implements these approximate derivatives? How would the kernels look like if we instead want to filter the image using convolution? In both cases, where is the center pixel in the filter kernel? Why does it not matter if we use correlation or convolution for filters like Gaussian or mean?

- 1.2. (1 point) Based upon the linear separability of the Prewitt and Sobel filters (see lecture slides), reinterpret the effect of the Prewitt and Sobel filters on images.

Deliverables: Write the two filters for y -derivatives in their separable forms. Explain in particular why they are more likely to be robust to noise than the simple finite difference approximation of the previous question.

2. Segmentation by thresholding

- 2.1. (1 point) Perform foreground-background segmentation using intensity thresholding on the `pillsect.png` image separating the objects from the background. Start by converting the image to grayscale, then apply thresholding (e.g. using `np.where`) using the threshold value 100 or 100/256, depending on whether you represent the grayscale image in unsigned 8 bit or floating point.

Deliverables: Show the original image, grayscale image and your binary segmentation image side-by-side, and include the essential code snippet in the

report. Did we succeed in separating the foreground objects from the background?

- 2.2. (1 point) Plot the intensity histogram of the grayscale image of `pillsect.png` from the previous question (you can for instance use `matplotlib.pyplot.hist`). Was the choice of threshold value in the previous question appropriate? What happens in the two cases where we either lower or increase the threshold value?
Deliverables: Include histogram plot in the report and your brief answer to the questions

3. **Fourier Transform – Theory** The following investigate theoretical properties of Fourier series and transform.

- 3.1. (1 point) What is the difference between a Fourier series and the Fourier Transform?
3.2. (2 point) Derive the continuous Fourier transform of $\delta(x - d) + \delta(x + d)$ for $x \in \mathbb{R}$ and some constant $d \in \mathbb{R}$.

Deliverables: Include crucial steps in derivations, and a short comment to each answer.

4. **Fourier Transform – Practice**

- 4.1. (1 point) The Fourier transform of a Gaussian function is also a Gaussian function but with a reciprocal relations for σ between the space and Fourier domain versions of the function. Using this fact and the convolution theorem, write a function `scale_fft`, which implements convolution with a Gaussian kernel in Fourier domain by constructing the filter kernel in the frequency space (i.e. without application of the discrete Fourier transformation on the kernel). The Gaussian kernel should be parameterised by its standard deviation (its scale) σ in space domain. Apply it to `trui.png` for a range of σ scales of your choice.

Deliverables: Include the code for your function in the report and explain how convolution can be implemented using the Fourier transform. Include illustrations of the function applied to `trui.png` at different σ scales.